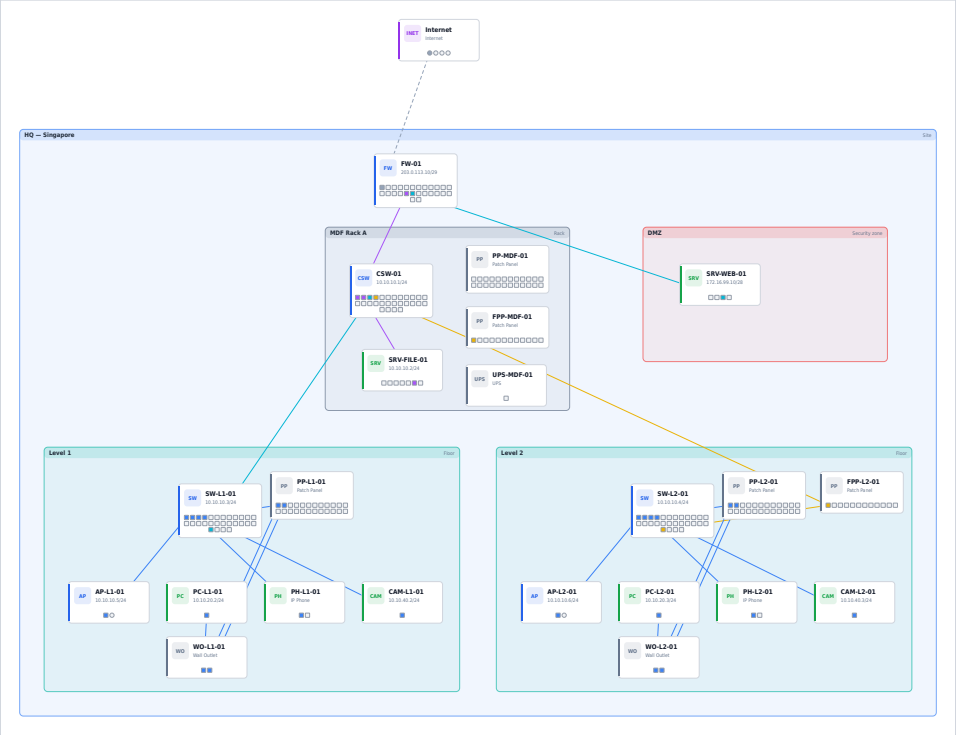
CORESCALE

Network design

NETWORK DESIGN REPORT

Corescale HQ

Head-office LAN: dual-floor access layer, DMZ, and server rack.



Client	**Corescale**
Prepared by	**Corescale**
Revision	**A**
Date	**20 October 2026**

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FW-01 — Campus firewall 16×1G, 8×SFP+, 2×QSFP28 11

SW-L1-01 — 24-port PoE+ access switch, 4×SFP+ (370 W) 12

SW-L2-01 — 24-port PoE+ access switch, 4×SFP+ (370 W) 12

FPP-L2-01 — 12-port duplex LC fibre patch panel 13

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PP-L1-01 — 24-port Cat6A patch panel 13

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1 Summary

Head-office LAN: dual-floor access layer, DMZ, and server rack.

Client	Corescale
Sites	HQ — Singapore
Devices	23 — 6 network, 8 end devices, 8 infrastructure, 1 cloud/WAN
Cables	21 cables in 16 runs
VLANs	5
Subnets	9
Wireless networks	2
Racks	1

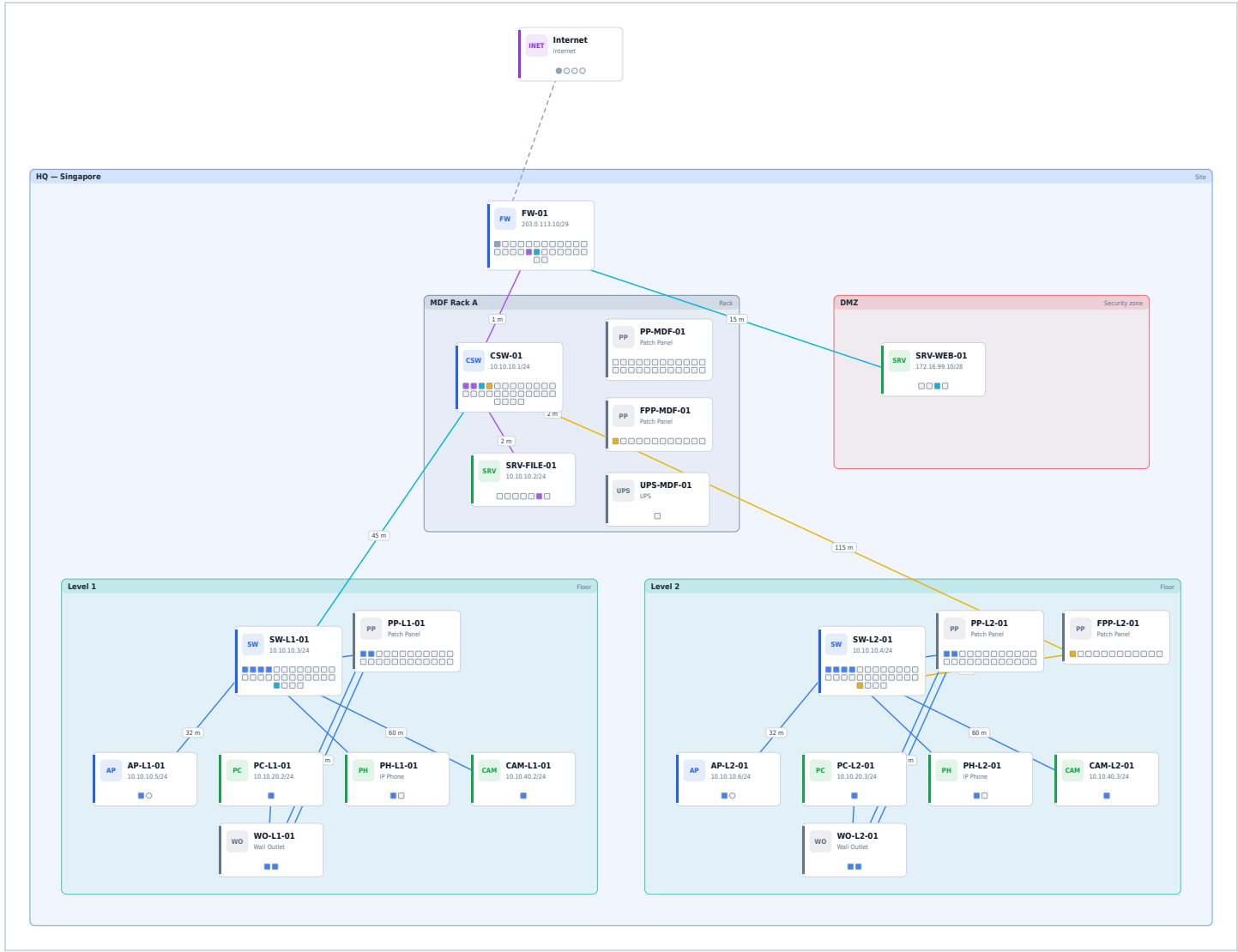
Design check result

No errors or warnings. 0 informational note(s).

Devices by type

Type	Count
Firewall / UTM	1
L2 Switch	2
L3 Switch	1
Wi-Fi AP	2
IP Camera	2
IP Phone	2
PC	2
Server	2
Patch Panel	5
UPS	1
Wall Outlet	2
Internet	1

2 Topology



Logical / WAN DAC Multimode fibre Single-mode fibre Copper

3 Device inventory

Name	Type	Model	Location	Rack	Address	Ports
AP-L1-01	Wi-Fi AP	Wi-Fi 6 AP 2×2, 1G uplink (PoE+)	Level 1	—	10.10.10.5/24	1/2
AP-L2-01	Wi-Fi AP	Wi-Fi 6 AP 2×2, 1G uplink (PoE+)	Level 2	—	10.10.10.6/24	1/2
CSW-01	L3 Switch	Distribution switch 24×SFP+, 4×QSFP+	HQ — Singapore	MDF Rack A U42	10.10.10.1/24	4/28
FW-01	Firewall / UTM	Campus firewall 16×1G, 8×SFP+, 2×QSFP28	HQ — Singapore	—	203.0.113.10/29	3/26
SW-L1-01	L2 Switch	24-port PoE+ access switch, 4×SFP+ (370 W)	Level 1	—	10.10.10.3/24	5/28
SW-L2-01	L2 Switch	24-port PoE+ access switch, 4×SFP+ (370 W)	Level 2	—	10.10.10.4/24	5/28
CAM-L1-01	IP Camera	Fixed dome camera (802.3af)	Level 1	—	10.10.40.2/24	1/1
CAM-L2-01	IP Camera	Fixed dome camera (802.3af)	Level 2	—	10.10.40.3/24	1/1
PC-L1-01	PC	—	Level 1	—	10.10.20.2/24	1/1
PC-L2-01	PC	—	Level 2	—	10.10.20.3/24	1/1
PH-L1-01	IP Phone	Desk IP phone with PC passthrough (802.3af)	Level 1	—	–	1/2
PH-L2-01	IP Phone	Desk IP phone with PC passthrough (802.3af)	Level 2	—	–	1/2
SRV-FILE-01	Server	2U server, 4×1G + 2×25G + iDRAC/iLO	HQ — Singapore	MDF Rack A U1–U2	10.10.10.2/24	1/7
SRV-WEB-01	Server	1U server, 2×1G + 2×25G	DMZ	—	172.16.99.10/28	1/4
FPP-L2-01	Patch Panel	12-port duplex LC fibre patch panel	Level 2	—	–	1/12
FPP-MDF-01	Patch Panel	12-port duplex LC fibre patch panel	HQ — Singapore	MDF Rack A U40	–	1/12
PP-L1-01	Patch Panel	24-port Cat6A patch panel	Level 1	—	–	2/24
PP-L2-01	Patch Panel	24-port Cat6A patch panel	Level 2	—	–	2/24
PP-MDF-01	Patch Panel	24-port Cat6A patch panel	HQ — Singapore	MDF Rack A U41	–	0/24
UPS-MDF-01	UPS	Line-interactive UPS 1500 VA / 1000 W	HQ — Singapore	MDF Rack A U3–U4	–	0/1
WO-L1-01	Wall Outlet	—	Level 1	—	–	2/2
WO-L2-01	Wall Outlet	—	Level 2	—	–	2/2

4 IP addressing

Subnets

Name	Network	VLAN	Gateway	DHCP range	Used
HQ block	10.10.0.0/16	—	—	—	1540/65534 (2%)
Core transit	10.10.0.0/30	—	—	—	2/2 (100%)
Management	10.10.10.0/24	10 · Management	10.10.10.1	—	6/254 (2%)
Staff	10.10.20.0/24	20 · Staff	10.10.20.1	10.10.20.128 – 10.10.20.254	130/254 (51%)
Voice	10.10.30.0/24	30 · Voice	10.10.30.1	10.10.30.128 – 10.10.30.254	128/254 (50%)
CCTV	10.10.40.0/24	40 · CCTV	10.10.40.1	—	3/254 (1%)
Guest	10.10.50.0/23	50 · Guest	10.10.50.1	10.10.51.0 – 10.10.51.254	256/510 (50%)
DMZ	172.16.99.0/28	—	—	—	2/14 (14%)
ISP handoff	203.0.113.8/29	—	—	—	1/6 (17%)

Interface addresses

Address	Device	Interface	Subnet	Type
10.10.0.1/30	**FW-01**	x1	Core transit	Static
10.10.0.2/30	**CSW-01**	xe1	Core transit	Static
10.10.10.1/24	**CSW-01**	Vlan10	Management	Static
10.10.10.2/24	**SRV-FILE-01**	ens1	Management	Static
10.10.10.3/24	**SW-L1-01**	Vlan10	Management	Static
10.10.10.4/24	**SW-L2-01**	Vlan10	Management	Static
10.10.10.5/24	**AP-L1-01**	eth0	Management	Static
10.10.10.6/24	**AP-L2-01**	eth0	Management	Static
10.10.20.1/24	**CSW-01**	Vlan20	Staff	Static
10.10.20.2/24	**PC-L1-01**	eth1	Staff	Static
10.10.20.3/24	**PC-L2-01**	eth1	Staff	Static
10.10.30.1/24	**CSW-01**	Vlan30	Voice	Static
10.10.40.1/24	**CSW-01**	Vlan40	CCTV	Static
10.10.40.2/24	**CAM-L1-01**	eth0	CCTV	Static
10.10.40.3/24	**CAM-L2-01**	eth0	CCTV	Static
10.10.50.1/23	**CSW-01**	Vlan50	Guest	Static
172.16.99.1/28	**FW-01**	x2	DMZ	Static
172.16.99.10/28	**SRV-WEB-01**	ens1	DMZ	Static
203.0.113.10/29	**FW-01**	port1	ISP handoff	Static

5 VLANs and Wi-Fi

VLANs

Tag	Name	Purpose	Subnet	Access	Trunks
10	**Management**	Network device and server management	10.10.10.0/24	1	6
20	**Staff**	Wired and wireless staff clients	10.10.20.0/24	2	6
30	**Voice**	IP phones	10.10.30.0/24	2	4
40	**CCTV**	IP cameras and NVR	10.10.40.0/24	2	4
50	**Guest**	Internet-only guest Wi-Fi	10.10.50.0/23	0	6

Wireless networks

SSID	VLAN	Security	Bands	Visibility
Corescale	20 · Staff	WPA3-Enterprise	5 GHz, 6 GHz	Broadcast
Corescale-Guest	50 · Guest	Enhanced Open (OWE)	2.4 GHz, 5 GHz	Broadcast

Served by 2 access point(s).

6 Routing and services

Network services

Domain	hq.corescale.local
DNS servers	1.1.1.1, 9.9.9.9
NTP servers	pool.ntp.org
Syslog	—

Static routes

Device	Destination	Next hop	Interface	AD	Description
FW-01	10.10.0.0/16	10.10.0.2	—	1	Internal networks via core
CSW-01	0.0.0.0/0	10.10.0.1	—	1	Default via firewall

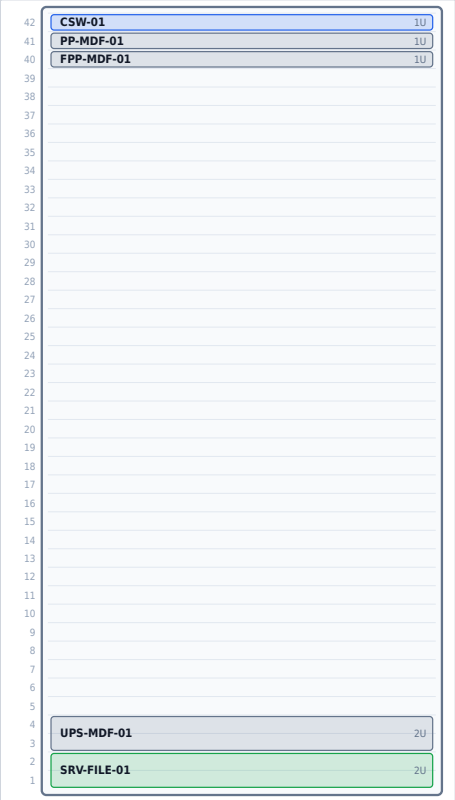
NAT

Device	Type	Inside	Outside	Service	Description
FW-01	Source PAT	10.10.0.0/16	port1 (203.0.113.10)	any	Internet access
FW-01	Port forward	172.16.99.10	203.0.113.11	tcp/443	Public website

7 Racks and power

MDF Rack A (42U)

7 of 42U used · 750 W load on 1000 W of UPS capacity (75%)



Position	Device	Model	Height	Power
U42	CSW-01	Distribution switch 24×SFP+, 4×QSFP+	1U	150 W
U41	PP-MDF-01	24-port Cat6A patch panel	1U	—
U40	FPP-MDF-01	12-port duplex LC fibre patch panel	1U	—
U3–U4	UPS-MDF-01	Line-interactive UPS 1500 VA / 1000 W	2U	—
U1–U2	SRV-FILE-01	2U server, 4×1G + 2×25G + iDRAC/iLO	2U	600 W

PoE budgets

Switch	Powered devices	Demand	Budget	Status
SW-L1-01	3	30 W	370 W	OK
SW-L2-01	3	30 W	370 W	OK

8 Patch schedule

Every cable run end to end. Runs through patch panels and wall outlets list the panel ports they pass; lengths are the sum of all cables in the run.

Run	From	Via	To	Cables	Length	Service	Labels
C-001	CSW-01:xe1	—	FW-01:x1	DAC 1 m	1 m	Routed	–
C-002	CSW-01:xe2	—	SRV-FILE-01:ens1	DAC 2 m	2 m	10 · Management	–
C-003	CSW-01:xe3	—	SW-L1-01:xe1	OM4 MMF 45 m	45 m	Trunk (all, native 10)	–
C-004	CSW-01:xe4	FPP-MDF-01:1, FPP-L2-01:1	SW-L2-01:xe1	OS2 SMF 2 m + OS2 SMF 115 m + OS2 SMF 3 m	120 m	Trunk (all, native 10)	–
C-005	FW-01:port1	—	Internet:link1	Logical / WAN	—	Routed	–
C-006	FW-01:x2	—	SRV-WEB-01:ens1	OM4 MMF 15 m	15 m	Routed	–
C-007	SW-L1-01:ge1	—	AP-L1-01:eth0	Cat6A 32 m	32 m	Trunk (10,20,50, native 10)	–
C-008	SW-L1-01:ge2	PP-L1-01:1, WO-L1-01:A	PC-L1-01:eth1	Cat6 1 m + Cat6A 42 m + Cat6 3 m	46 m	20 · Staff	–
C-009	SW-L1-01:ge3	—	PH-L1-01:lan0	Cat6	—	30 · Voice	–
C-010	SW-L1-01:ge4	—	CAM-L1-01:eth0	Cat6 60 m	60 m	40 · CCTV	–
C-011	SW-L2-01:ge1	—	AP-L2-01:eth0	Cat6A 32 m	32 m	Trunk (10,20,50, native 10)	–
C-012	SW-L2-01:ge2	PP-L2-01:1, WO-L2-01:A	PC-L2-01:eth1	Cat6 1 m + Cat6A 42 m + Cat6 3 m	46 m	20 · Staff	–
C-013	SW-L2-01:ge3	—	PH-L2-01:lan0	Cat6	—	30 · Voice	–
C-014	SW-L2-01:ge4	—	CAM-L2-01:eth0	Cat6 60 m	60 m	40 · CCTV	–
C-015	PP-L1-01:2	—	WO-L1-01:B (spare)	Cat6A 42 m	42 m	—	–
C-016	PP-L2-01:2	—	WO-L2-01:B (spare)	Cat6A 42 m	42 m	—	–

9 Port maps

CSW-01 — Distribution switch 24×SFP+, 4×QSFP+

4 of 28 ports in use · MDF Rack A U42

Port	State	Media	Speed	Mode / VLAN	Connected to	Via	Cables	Labels	PoE
xe1	Connected	SFP+	10G	Routed	FW-01:x1		DAC 1 m		
xe2	Connected	SFP+	10G	10 · Management	SRV-FILE-01:ens1		DAC 2 m		
xe3	Connected	10GBASE-SR	10G	Trunk (all, native 10)	SW-L1-01:xe1		OM4 MMF 45 m		
xe4	Connected	10GBASE-LR	10G	Trunk (all, native 10)	SW-L2-01:xe1	FPP-MDF-01:1, FPP-L2-01:1	OS2 SMF 2 m + OS2 SMF 115 m + OS2 SMF 3 m		
xe5–xe24	Free (20 ports)	SFP+	10G						
et1–et4	Free (4 ports)	QSFP+	40G						

FW-01 — Campus firewall 16×1G, 8×SFP+, 2×QSFP28

3 of 26 ports in use · —

Port	State	Media	Speed	Mode / VLAN	Connected to	Via	Cables	Labels	PoE
port1	Connected	RJ45	1G	Routed	Internet:link1		Logical / WAN		
port2–port16	Free (15 ports)	RJ45	1G						
x1	Connected	SFP+	10G	Routed	CSW-01:xe1		DAC 1 m		
x2	Connected	10GBASE-SR	10G	Routed	SRV-WEB-01:ens1		OM4 MMF 15 m		
x3–x8	Free (6 ports)	SFP+	10G						
hu1–hu2	Free (2 ports)	QSFP28	100G						

SW-L1-01 — 24-port PoE+ access switch, 4×SFP+ (370 W)

5 of 28 ports in use · —

Port	State	Media	Speed	Mode / VLAN	Connected to	Via	Cables	Labels	PoE
ge1	Connected	RJ45	1G	Trunk (10,20,50, native 10)	AP-L1-01:eth0		Cat6A 32 m		PSE at
ge2	Connected	RJ45	1G	20 · Staff	PC-L1-01:eth1	PP-L1-01:1, WO-L1-01:A	Cat6 1 m + Cat6A 42 m + Cat6 3 m		PSE at
ge3	Connected	RJ45	1G	30 · Voice	PH-L1-01:lan0		Cat6		PSE at
ge4	Connected	RJ45	1G	40 · CCTV	CAM-L1-01:eth0		Cat6 60 m		PSE at
ge5-ge24	Free (20 ports)	RJ45	1G						PSE at
xe1	Connected	10GBASE-SR	10G	Trunk (all, native 10)	CSW-01:xe3		OM4 MMF 45 m		
xe2-xe4	Free (3 ports)	SFP+	10G						

SW-L2-01 — 24-port PoE+ access switch, 4×SFP+ (370 W)

5 of 28 ports in use · —

Port	State	Media	Speed	Mode / VLAN	Connected to	Via	Cables	Labels	PoE
ge1	Connected	RJ45	1G	Trunk (10,20,50, native 10)	AP-L2-01:eth0		Cat6A 32 m		PSE at
ge2	Connected	RJ45	1G	20 · Staff	PC-L2-01:eth1	PP-L2-01:1, WO-L2-01:A	Cat6 1 m + Cat6A 42 m + Cat6 3 m		PSE at
ge3	Connected	RJ45	1G	30 · Voice	PH-L2-01:lan0		Cat6		PSE at
ge4	Connected	RJ45	1G	40 · CCTV	CAM-L2-01:eth0		Cat6 60 m		PSE at
ge5-ge24	Free (20 ports)	RJ45	1G						PSE at
xe1	Connected	10GBASE-LR	10G	Trunk (all, native 10)	CSW-01:xe4	FPP-L2-01:1, FPP-MDF-01:1	OS2 SMF 3 m + OS2 SMF 115 m + OS2 SMF 2 m		
xe2-xe4	Free (3 ports)	SFP+	10G						

FPP-L2-01 — 12-port duplex LC fibre patch panel

1 of 12 ports in use · —

Port	State	Media	Speed	Mode / VLAN	Connected to	Via	Cables	Labels	PoE
1	Connected	LC	400G		CSW-01:xe4 ↔ SW-L2-01:xe1	FPP-MDF-01:1	OS2 SMF 2 m + OS2 SMF 115 m + OS2 SMF 3 m		
2–12	Free (11 ports)	LC	400G						

FPP-MDF-01 — 12-port duplex LC fibre patch panel

1 of 12 ports in use · MDF Rack A U40

Port	State	Media	Speed	Mode / VLAN	Connected to	Via	Cables	Labels	PoE
1	Connected	LC	400G		CSW-01:xe4 ↔ SW-L2-01:xe1	FPP-L2-01:1	OS2 SMF 2 m + OS2 SMF 115 m + OS2 SMF 3 m		
2–12	Free (11 ports)	LC	400G						

PP-L1-01 — 24-port Cat6A patch panel

2 of 24 ports in use · —

Port	State	Media	Speed	Mode / VLAN	Connected to	Via	Cables	Labels	PoE
1	Connected	RJ45	10G		SW-L1-01:ge2 ↔ PC-L1-01:eth1	WO-L1-01:A	Cat6 1 m + Cat6A 42 m + Cat6 3 m		
2	One side cabled	RJ45	10G		WO-L1-01:B		Cat6A 42 m		
3–24	Free (22 ports)	RJ45	10G						

PP-L2-01 — 24-port Cat6A patch panel

2 of 24 ports in use · —

Port	State	Media	Speed	Mode / VLAN	Connected to	Via	Cables	Labels	PoE
1	Connected	RJ45	10G		SW-L2-01:ge2 ↔ PC-L2-01:eth1	WO-L2-01:A	Cat6 1 m + Cat6A 42 m + Cat6 3 m		
2	One side cabled	RJ45	10G		WO-L2-01:B		Cat6A 42 m		
3–24	Free (22 ports)	RJ45	10G						

10 Bill of materials

Section	Item	Qty	Order	Notes
Hardware	12-port duplex LC fibre patch panel	**2**		FPP-MDF-01, FPP-L2-01
	1U server, 2×1G + 2×25G	**1**		SRV-WEB-01
	24-port Cat6A patch panel	**3**		PP-MDF-01, PP-L1-01, PP-L2-01
	24-port PoE+ access switch, 4×SFP+ (370 W)	**2**		SW-L1-01, SW-L2-01
	2U server, 4×1G + 2×25G + iDRAC/iLO	**1**		SRV-FILE-01
	Campus firewall 16×1G, 8×SFP+, 2×QSFP28	**1**		FW-01
	Desk IP phone with PC passthrough (802.3af)	**2**		PH-L1-01, PH-L2-01
	Distribution switch 24×SFP+, 4×QSFP+	**1**		CSW-01
	Fixed dome camera (802.3af)	**2**		CAM-L1-01, CAM-L2-01
	Line-interactive UPS 1500 VA / 1000 W	**1**		UPS-MDF-01
	Wi-Fi 6 AP 2×2, 1G uplink (PoE+)	**2**		AP-L1-01, AP-L2-01
	PC (model not chosen)	**2**		PC-L1-01, PC-L2-01
	Wall Outlet (model not chosen)	**2**		WO-L1-01, WO-L2-01
Optics	SFP+ 10GBASE-LR	**2**		10G · SMF 1310 nm · Duplex LC
	SFP+ 10GBASE-SR	**4**		10G · MMF 850 nm · Duplex LC
Cabling	Cat6	**8**	141 m	128 m measured + 10% slack; 2 without a length
	Cat6A	**6**	256 m	232 m measured + 10% slack
	DAC	**2**	4 m	3 m measured + 10% slack
	OM4 MMF	**2**	66 m	60 m measured + 10% slack
	OS2 SMF	**3**	132 m	120 m measured + 10% slack
Racks	42U rack	**1**		MDF Rack A

Cable quantities count one cable per link and add 10% slack to measured lengths. Links without a length need measuring before ordering.

11 Design check

The design passes every check: addressing, VLANs, Wi-Fi, names, topology, cabling, power and racks.